

MAE 146: Intro to ML with Python

Homework #1

Assigned: April 2, 2026

Due: April 16, 2026

Corrections: April 23, 2026

Instructions: There are two parts to this homework: Analytical and Programming, which will be submitted separately on Gradescope. Please see the two sections below for instructions on submission.

Analytical Problems (24 points)

This homework is a review of math fundamentals that are important for this course. (If you are not comfortable with these concepts, you may have difficulties in 146.) For these problems you should upload solutions that you wrote by hand.

Homework exercises:

1. **(4 points)** Calculate $\lim_{x \rightarrow -1} f(x)$, $\lim_{x \rightarrow +1} f(x)$ for the functions below. Then obtain the intervals where f is differentiable and provide its derivative.
 - (a) $f(x) = \max\{0, x\}$.
 - (b) $f(x) = \frac{1}{1 + e^{-x}}$.
2. **(4 points)** Calculate the lengths of the vectors $a = [0.4, 0.3]^\top$, and $b = [-0.15, 2]^\top$. What is the angle between a and b in radians? Given two vectors $x_1 = [a_1, a_2, a_3]^\top$ and $x_2 = [a_1, -a_2, a_3]^\top$, provide a formula for when these vectors are orthogonal in terms of their entries.
3. **(3 points)**
 - (a) Given any vector x , write down an expression for a unit vector pointing in the same direction as x (use the norm of the vector).
 - (b) Recall from linear algebra the projection of one vector onto another. Consider now $x_1 = [a_1, a_2, a_3]^\top$ and $x_2 = [a_1, -a_2, a_3]^\top$. In what direction is the projection of x_1 onto x_2 ? Find an expression for the number c such that the projection of x_1 onto x_2 is given by cu where u is a unit vector in the direction of the said projection.
4. **(3 points)**

- (a) Given a fixed d -dimensional vector θ and a scalar offset θ_0 describing a hyperplane $\pi : \theta \cdot x + \theta_0 = 0$ (i.e., where θ is perpendicular to the hyperplane, and x is a point in space), how many other alternative values of θ' and θ'_0 can be used to describe the same π ?
- (b) Provide a simple check to verify that a vector x is orthogonal to a plane π characterized by θ and θ_0 .
- (c) Given a point x in the n -dimensional space, find the signed distance between x and the hyperplane π . (This is equal to the perpendicular distance between the hyperplane and x , and is positive when x is on the same side of the plane as θ , negative otherwise.)

5. (2 points)

- (a) Find an expression for the orthogonal projection of a point x onto a plane π that is characterized by θ and θ_0 . (Hint: you may be able to use some of your previous results here.)
- (b) Consider a hyperplane in a d -dimensional space. If we project a point onto the plane, can we recover the original point from this projection?

6. (4 points) Given a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, with argument $\theta = (\theta_1, \theta_2, \dots, \theta_n)$ and value $f(\theta_1, \theta_2, \dots, \theta_n)$, we denote its gradient function by $\nabla_\theta f : \mathbb{R}^d \rightarrow \mathbb{R}^d$, taking values

$$\nabla_\theta f(\theta) = \left[\frac{\partial f}{\partial \theta_1}(\theta), \frac{\partial f}{\partial \theta_2}(\theta), \dots, \frac{\partial f}{\partial \theta_d}(\theta) \right]^\top.$$

Consider $f(\theta) = \theta_1^2 + \theta_2^2$ and compute the gradient ∇f . Visualize $f(\theta)$ as a surface over the (θ_1, θ_2) plane, what is the shape of a curve $f(\theta) = K$ for a fixed real number $K > 0$? Consider the level curves $\theta_1^2 + \theta_2^2 = K$ as K increases from 0 to $+\infty$. Does the graph (surface) of $f(\theta)$ have a global maximum, or global minimum, or neither? Using the gradient of f , obtain a direction in the (θ_1, θ_2) plane along which we observe a decrease of $f(\theta)$.

7. (4 points) Consider the matrices:

$$B = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 4 & 4 \\ 5 & 6 & 4 \end{bmatrix}$$

- (a) Determine the rank of B .
- (b) Can we compute B^{-1} ? If so, what is the product BB^{-1} ?
- (c) Suppose $u = [1, 3]^\top$ and $v = [-1, 1]^\top$. Find the inner product $u^\top v$ of u and v (also called dot product, and sometimes denoted as $\langle u, v \rangle$ or as $u \cdot v$).
- (d) For the same u and v , find the outer product $uv^\top$.

Programming Exercises (24 points)

Instructions:

Please fill out the file `main HW1.py` in your assignment to answer the following questions. (Make sure that the file you upload is named exactly as `main HW1.py` and the function names are the same as the ones given in the template).

Problems

1. **(4 points)** Write a function called `randomization` that takes in a positive integer n and returns A , a random $n \times 1$ numpy array. Use the format for the function `randomization` available in `main.py`.
2. **(4 points)** Write a function called `operations` that takes as input two positive integers h and w , makes two random matrices A and B , of size $h \times w$, and returns A , B , and s , the sum of A and B . Use the format for the function `operations` available in `main.py`.
3. **(4 points)** Write a function called `norm` that takes as input two Numpy column arrays A and B , adds them, and returns s , the 2 norm of their sum. Use the format for the function `norm` available in `main.py`.
4. **(4 points)** A simple neural network is described by two inputs x_1, x_2 , and one output z_1 . The output z_1 depends on the inputs, and the importance of the inputs is modeled through the weights w_{11}, w_{21} . In particular, w_{11} is the weight of input 1 on output 1, and w_{21} is the weight of input 2 on output 1. In general, w_{ji} represents the weight of input j on output i .

The output is given exactly as $z_1 = f(w_{11}x_1 + w_{21}x_2)$, where f will be a nonlinear function such as the hyperbolic tangent $\tanh$. If we express our inputs and weights as matrices:

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad w = \begin{bmatrix} w_{11} \\ w_{21} \end{bmatrix}$$

we can just express $z_1 = \tanh(w^\top x)$. Write a function called `neural network` which will apply a neural network operation with 2 inputs and 1 output and a given weight matrix. Use the format for the function `neural network` available in `main.py`.

5. **(4 points)** Implement the following function in python:

$$f(x, y) = \begin{cases} x \cdot y & \text{if } x \leq y \\ x/y & \text{else.} \end{cases}$$

Note that x, y are scalars.

6. **(4 points)** Now write a vector function `vector_function`, which will apply the operation $f(x, y)$ defined in the previous problem element- wisely with input vectors with same dimension x and y .

Hint: A scalar function can only handle scalar inputs. We can use the function `np.vectorize()` to turn it into a vectorized function. Note that the input argument of `np.vectorize()` should be a scalar function, and the output of `np.vectorize()` is a new function that can handle input vectors.